MCO-1D

Operator Manual

Flight operations, controls, and procedures

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1. Flight Envelope

The MCO-1D flight envelope is measured on first flight using H-PERF-6. Until that measurement is complete, HOVER_CAPABLE = None and the system treats both hover and forward-flight modes as available. After the first flight the operator should confirm the result using set_hover_capable.

Envelope determination — first flight procedure

1. Launch to 10m AGL. Establish stable cruise at 9 m/s.
2. Reduce speed in 0.5 m/s steps. Hold each speed for 10 seconds.
3. Monitor VSI. If VSI drops below –0.5 m/s, the aircraft is losing altitude — this is V_min. Return to cruise.
4. Attempt a hover at V_min: slow further toward 0 m/s with lever at trim detent.
5. If aircraft maintains altitude with V < 1.0 m/s for 10s: hover capable.
6. After landing: GS command set_hover_capable true or false.
7. The ALC post-flight debrief will report the full envelope measurement.

2. Lever Map

Range	Zone	Description
100–88%	**MAX_THRUST**	Full sprint, maximum frequency and amplitude
88–78%	**HIGH_THRUST**	Powered climb, jump manoeuvres
78–68%	**POWERED_CLIMB**	Climb zone, whisper improves efficiency here
68–55%	**CRUISE**	Normal cruise, discovery most active
55–45%	**TRIM_DETENT**	Hover / slow cruise hold
45–38%	**APPROACH**	Slow approach configuration
38–28%	**GLIDE_SOAR**	Near-unpowered glide, thermals, regen active
28–16%	**DESCENT**	Pigeon descent, dihedral braking, regen active
16–6%	**ACTIVE_BRAKE**	Cascade reversal braking, regen active
6–0%	**EMERGENCY_STOP**	Maximum braking cascade 160–180°, regen active

3. Control Mapping (key entries)

Control input	Hypothesis	Result
Both pedals + roll (hover)	`H-OV-2`	Pure lateral translation
Aft pitch >0.5 + roll + both pedals	`H-OV-3 Mode A`	Diagonal translation
Roll + both pedals + pedal diff	`H-OV-3 Mode B`	Arced lateral
Aft pitch >0.5 + roll + pedals + diff	`H-OV-3 Mode C`	Full envelope
Aft pitch >0.5 + both pedals (hover)	`H-RC-2`	Rearward flight
High bank + aft pitch	`H-PULL-1`	Pitch authority Modes A/B/C
Bank >135° + aft pitch	`H-PULL-1 Mode C`	Inverted pull → positive G
Lever 0–6%	`Emergency stop`	Cascade 160–180°, max regen

4. Forward-Flight SAR Procedures

When HOVER_CAPABLE = False (or as an endurance-efficient alternative), the aircraft conducts SAR missions in forward-flight mode. The ALC will automatically gate hover hypotheses and route to forward-flight equivalents.

Slow orbit (H-SAR-FF-1)

Fly a banked circle at V_min centred on the target. 15° bank at V_min=4 m/s gives 6.1m orbit radius. Nose 8–12° down keeps sensor on target. Pilot flies the circle; genome optimises the kinematics.

Area search (H-SAR-FF-2)

Fly parallel tracks at V_min with direction reversals at track ends. Track spacing set by sensor FOV and altitude. The genome optimises the reversal turn kinematics (note: pattern execution is autopilot-deferred).

Sensor approach (H-SAR-FF-3)

Descend in a spiral orbit while keeping target in sensor FOV. Nose-down pitch increases as altitude decreases. Transitions to H-PREC-2 short-field landing at 10m AGL.

Payload drop (H-SAR-FF-4)

Minimum-speed pass at lowest safe altitude. Release at closest approach. Per-wing amplitude reduction prevents post-release climb surge.

Short-field landing (H-PREC-2)

Approach at V_min. Flare at 2.5m AGL with dihedral braking. Aim: <2 m/s horizontal, <1 m/s vertical at touchdown.

5. ALC Communication

The ALC follows the sterile cockpit protocol. During flight it is nearly always silent. The primary communication events are the pre-flight briefing and post-flight debrief.

Command	When	Purpose
alc_preflight_brief	Before flight	What was learned last session, what hypotheses are queued
alc_debrief	After landing	Full session narrative: evaluations, genome shifts, regen, wear
learning_rate good	Any time	Positive genome reinforcement for current configuration
learning_rate poor	Any time	Negative genome reinforcement for current configuration
alc_ack_alert	When tier-3 fires	Acknowledge safety alert
wear_status	Post-flight	Per-wing mechanical health and stroke counts
regen_status	Post-flight	Session and all-time regen energy totals

6. Regenerative Energy Recovery

Regen activates automatically whenever the lever is below the approach zone. No operator action required. Monitor with regen_status after flight. Enable on ODrive with regen_config_advice command.

Note: total regen per flight is small (100–500 mWh typically) but cumulative across many flights is meaningful. The stewardship value — never wasting energy as heat when it can be reclaimed — is the primary rationale.